

energy



Name:

Date:

10 The diagram shows the reaction pathway for a reversible reaction.

Which statements are correct?

- 1 The enthalpy change for the backward reaction is $+90 \text{ kJ mol}^{-1}$.
- 2 The forward reaction is exothermic.
- 3 The enthalpy change for the forward reaction is -30 kJ mol^{-1} .

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

Correct Answer: A B C D

Explanation:

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6 Methanol may be prepared by the reaction between carbon monoxide and hydrogen.

$$\text{CO(g)} + 2\text{H}_2\text{(g)} \rightarrow \text{CH}_3\text{OH(g)}$$

The relevant average bond energies are given below.

$E(\text{C}=\text{O})$	1077 kJ mol^{-1}
$E(\text{C}-\text{O})$	360 kJ mol^{-1}
$E(\text{C}-\text{H})$	410 kJ mol^{-1}
$E(\text{H}-\text{H})$	436 kJ mol^{-1}
$E(\text{O}-\text{H})$	460 kJ mol^{-1}

What is the enthalpy change of this reaction?

A -537 kJ mol^{-1}
B -101 kJ mol^{-1}
C $+101 \text{ kJ mol}^{-1}$
D $+537 \text{ kJ mol}^{-1}$

Correct Answer: A B C D

Explanation:

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1 Some bond energy values are listed below.

bond	bond energy / kJ mol^{-1}
C-H	410
C-Cl	340
Cl-Cl	244
Br-Br	193

These bond energy values relate to the following four reactions.

P $\text{Br}_2 \rightarrow 2\text{Br}$
Q $2\text{Cl} \rightarrow \text{Cl}_2$
R $\text{CH}_3 + \text{Cl} \rightarrow \text{CH}_3\text{Cl}$
S $\text{CH}_4 \rightarrow \text{CH}_3 + \text{H}$

What is the order of enthalpy changes of these reactions from most negative to most positive?

A $\text{P} \rightarrow \text{Q} \rightarrow \text{R} \rightarrow \text{S}$
B $\text{Q} \rightarrow \text{R} \rightarrow \text{S} \rightarrow \text{P}$
C $\text{R} \rightarrow \text{Q} \rightarrow \text{P} \rightarrow \text{S}$
D $\text{S} \rightarrow \text{P} \rightarrow \text{Q} \rightarrow \text{R}$

Correct Answer: A B C D

Explanation:

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The reaction of hydrogen gas with bromine gas to make hydrogen bromide is faster when the temperature is higher. This is because at a higher temperature:

A the reaction is more exothermic

B the gas molecules move with more energy

C the activation energy for the reaction is lower

D the energy needed to break the bonds in the molecules is less

Correct Answer: A B C D

Explanation:

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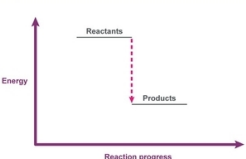
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The simple energy level diagram for a chemical reaction is shown. What does this diagram tell us about the reaction?



A it proceeds immediately at room temperature without heating

B it is a fast reaction but only starts at a high temperature

C it is endothermic

D it is exothermic

Correct Answer: A B C D

Explanation:

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Instant cold packs are used to treat sports injuries. One type of cold pack has a plastic bag with water inside. Inside the bag is a smaller bag containing solid ammonium nitrate. The outer bag is squeezed so that the inner bag bursts. The pack is shaken and quickly gets cold as the ammonium nitrate dissolves in the water. Why does the bag get cold?



A) The reaction is endothermic so heat energy is released into the surroundings

B) The reaction is exothermic so heat energy is released into the surroundings

C) The reaction is endothermic so heat energy is taken in from the surroundings

D) The reaction is exothermic so heat energy is taken in from the surroundings

Correct Answer: A B C D

Explanation:

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Which of these statements about exothermic reactions is correct:

A) Exothermic reactions transfer energy to the surroundings.

B) Exothermic reactions take in energy from the surroundings.

C) Exothermic reactions are energy neutral overall.

D) None of the above.

Correct Answer: A B C D

Explanation:

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Barium hydroxide reacts with ammonium chloride according to this word equation: *barium hydroxide + ammonium chloride → ammonia + water + barium chloride* The reaction is so endothermic that it can freeze water on the outside of a flask in which the reaction takes place. During this reaction bonds are being broken and bonds are being formed. Which one of the following statements is true?

- ☐ A The energy released making bonds in the products is smaller than the energy required to break bonds in the reactants.
- ☐ B The energy released making bonds in the products is greater than the energy required to break bonds in the reactants.
- ☐ C The energy released making bonds in the reactants is greater than the energy required to break bonds in the products.
- ☐ D The energy released making bonds in the reactants is less than the energy required to break bonds in the products.

Correct Answer: A B C D

Explanation:

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